

SLR Inference

Outline

- 1 From Estimation to Inference
- 2 Sampling Distribution of the OLS Estimators
- 3 Estimating the Error Variance
- 4 t-Test for the Slope
- 5 t-Test for the Intercept
- 6 Confidence Intervals for the Slope and Intercept
- 7 Partitioning Variability: The ANOVA Identity

From Estimation to Inference

Where We Left Off

- Fitted the simple linear regression model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$ by least squares
- Estimated $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
- Estimated $\hat{\sigma}^2 = MS_E = \frac{SS_E}{n-2} = \frac{\sum_{i=1}^n \hat{\epsilon}_i^2}{n-2} = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n-2}$

If we drew a different sample, would $\hat{\beta}_1$ come out the same?

Sampling Distribution of the OLS Estimators

Sampling Distribution of $\hat{\beta}_1$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \sum_{i=1}^n c_i Y_i, \quad c_i = \frac{x_i - \bar{x}}{S_{xx}}$$

- $\hat{\beta}_1$ is a linear combination of independent normal random variables, therefore, $\hat{\beta}_1$ is normally distributed

Further,

$$E[\hat{\beta}_1] = \beta_1 \quad (\text{unbiased})$$

$$\text{Var}(\hat{\beta}_1) = \sigma^2 \sum_i c_i^2 = \frac{\sigma^2}{S_{xx}}$$

Therefore:

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

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Sampling Distribution $\hat{\beta}_0$

Since $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$ and

$$E[\hat{\beta}_0] = \beta_0 \quad (\text{unbiased})$$

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$$

Therefore:

$$\hat{\beta}_0 \sim N \left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right) \right)$$

Estimating the Error Variance

σ^2 Is Unknown — Estimate It from the Residuals

$$\hat{\sigma}^2 = MSE = \frac{SSE}{n-2}, \quad SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- Two degrees of freedom are used up estimating β_0, β_1
- MSE is unbiased: $E[MSE] = \sigma^2$
- Key distributional result:

$$\frac{(n-2)MSE}{\sigma^2} \sim \chi_{n-2}^2$$

Standard Errors

- Replace the unknown σ^2 by MSE in the variance formulas

$$se(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} = \sqrt{\frac{MSE}{S_{xx}}}$$

$$se(\hat{\beta}_0) = \sqrt{\hat{\sigma}^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)} = \sqrt{MSE \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)}$$

- Because σ^2 is estimated, the reference distribution moves from normal to t_{n-2}

t-Test for the Slope

Test Statistic for β_1

$$T_0 = \frac{\hat{\beta}_1 - \beta_{1,0}}{se(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{MSE/S_{xx}}} \sim t_{n-2} \quad \text{under } H_0$$

- $H_0 : \beta_1 = \beta_{1,0}$ vs. $H_1 : \beta_1 \neq \beta_{1,0}$ (or one-sided)
- $n - 2$ degrees of freedom — one for each parameter already estimated

Decision Rule

- Two-sided test at level α : reject H_0 if $|t_0| > t_{\alpha/2, n-2}$
- Equivalently, reject if the p -value $< \alpha$
- Most common special case: $\beta_{1,0} = 0$
 - Tests whether x has *any* linear relationship with y
 - Called the **test for significance of regression**

Running Example

- Weekly advertising expenditure x (\$100s) and weekly sales y (\$1000s), $n = 20$

$\bar{x}$	$\bar{y}$	S_{xx}	S_{xy}	SST
10	50	200	300	500

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{300}{200} = 1.5$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 50 - 1.5(10) = 35$$

We carry these numbers through the rest of the lecture.

Testing $H_0 : \beta_1 = 0$ for the Example

$$SSE = SST - \hat{\beta}_1 S_{xy} = 500 - 1.5(300) = 50$$

$$MSE = \frac{SSE}{n - 2} = \frac{50}{18} = 2.778$$

$$se(\hat{\beta}_1) = \sqrt{\frac{2.778}{200}} = 0.118$$

$$t_0 = \frac{1.5 - 0}{0.118} = 12.72$$

- $t_{0.025,18} = 2.101$; since $|t_0| = 12.72 \gg 2.101$, reject H_0
- Strong evidence of a linear relationship between advertising and sales

t-Test for the Intercept

Test Statistic for β_0

$$T_0 = \frac{\hat{\beta}_0 - \beta_{0,0}}{se(\hat{\beta}_0)} \sim t_{n-2} \quad \text{under } H_0 : \beta_0 = \beta_{0,0}$$

- Same mechanics as the slope test, different standard error
- Testing $\beta_0 = 0$ is rarely of scientific interest unless $x = 0$ is meaningful

Question: for the advertising example, is $\beta_0 = 0$ an interesting hypothesis?

Confidence Intervals for the Slope and Intercept

CI for β_1

$$\hat{\beta}_1 \pm t_{\alpha/2, n-2} se(\hat{\beta}_1)$$

- A $100(1 - \alpha)\%$ CI: the range of slopes plausible given the data
- Narrower when S_{xx} is large or MSE is small

CI for β_1 — Example

$$\hat{\beta}_1 \pm t_{0.025, 18} se(\hat{\beta}_1)$$

$$1.5 \pm 2.101(0.118)$$

$$1.5 \pm 0.248$$

$$95\% \text{ CI : } (1.252, 1.748)$$

- Zero is nowhere near this interval — consistent with rejecting $H_0 : \beta_1 = 0$

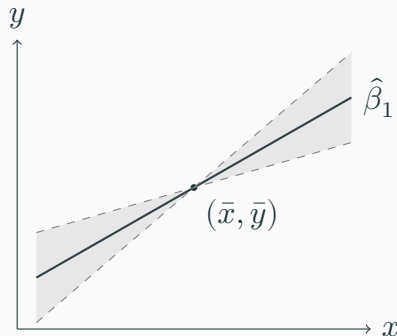
CI for β_0

$$\hat{\beta}_0 \pm t_{\alpha/2, n-2} se(\hat{\beta}_0)$$

- Same construction, using $se(\hat{\beta}_0) = \sqrt{MSE(1/n + \bar{x}^2/S_{xx})}$
- Interpretation only meaningful if $x = 0$ falls in or near the range of the data

Geometric Intuition: A Band of Plausible Lines

- A CI for β_1 corresponds to a range of *slopes* through $(\bar{x}, \bar{y})$
- Every line in the shaded wedge is compatible with the data at level α
- The fitted line is the single most plausible member of that family



Partitioning Variability: The ANOVA Identity

Splitting Total Variation

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{SST} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{SSR} + \underbrace{\sum_i (y_i - \hat{y}_i)^2}_{SSE}$$

- *SST*: total variability in the response
- *SSR*: variability *explained* by the regression on x
- *SSE*: variability left *unexplained* (residual)

Degrees of Freedom Split the Same Way

$$\underbrace{n-1}_{SST} = \underbrace{1}_{SSR} + \underbrace{n-2}_{SSE}$$

- One degree of freedom for the regression — a single slope parameter
- $n - 2$ left for error, matching what was used for MSE

The ANOVA Table

Source	df	SS	MS	F_0
Regression	1	SSR	$MSR = SSR/1$	MSR/MSE
Residual (Error)	$n - 2$	SSE	$MSE = SSE/(n - 2)$	
Total	$n - 1$	SST		

- Rows sum: the df 's add to $n - 1$, the SS 's add to SST

ANOVA Table for the Example

Source	df	SS	MS	F_0
Regression	1	450	450.0	162.0
Residual	18	50	2.778	
Total	19	500		

- $SSR = \hat{\beta}_1 S_{xy} = 1.5(300) = 450$
- $F_0 = 450/2.778 = 162.0$

The F-Test for Significance of Regression

F -Test

$$F_0 = \frac{MSR}{MSE} \sim F_{1, n-2} \quad \text{under } H_0 : \beta_1 = 0$$

- Reject H_0 if $F_0 > F_{\alpha, 1, n-2}$
- For the example: $F_{0.05, 1, 18} = 4.41$; since $F_0 = 162.0 \gg 4.41$, reject H_0

F -Test vs. t -Test: Same Question, Two Routes

Question: in simple linear regression, how do the t -test and F -test for $H_0 : \beta_1 = 0$ relate?

- Check against the numbers above: $t_0^2 = 12.72^2 = 161.8 \approx F_0 = 162.0$ (rounding)
- In *multiple* regression the F -test generalizes to several predictors at once; the simple t -test does not

Coefficient of Determination

R^2 : How Much Variation Is Explained

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}, \quad 0 \leq R^2 \leq 1$$

- Proportion of the total variability in y explained by the regression on x
- For the example: $R^2 = 450/500 = 0.90$ — 90% of the variability in sales is explained by advertising expenditure

R^2 and the Correlation Coefficient

- In simple linear regression only:

$$R^2 = r^2$$

- $r = \sqrt{R^2} = \sqrt{0.90} = 0.949$ (sign taken from $\hat{\beta}_1$)
- This equivalence is special to SLR — it does *not* hold once there is more than one predictor

Prediction

Two Different Questions

- Fix a value x_0 . Two distinct targets:
 - the *mean* response $E[Y | x_0] = \beta_0 + \beta_1 x_0$
 - a *single new* observation $Y_0 = \beta_0 + \beta_1 x_0 + \varepsilon_0$
- Both are estimated by the same point, $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$
- But they do *not* share the same uncertainty

Confidence Interval for the Mean Response

$$\text{Var}(\hat{y}_0) = \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$$

$$\hat{y}_0 \pm t_{\alpha/2, n-2} \sqrt{MSE \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}$$

- Narrowest at $x_0 = \bar{x}$, widening as x_0 moves away from $\bar{x}$

Prediction Interval for a New Observation

$$\text{Var}(Y_0 - \hat{y}_0) = \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)$$

$$\hat{y}_0 \pm t_{\alpha/2, n-2} \sqrt{MSE \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}$$

- The extra “1” accounts for the variability of ε_0 itself, on top of the uncertainty in estimating the line
- Always wider than the CI for the mean response

CI and PI at $x_0 = 12$: The Example

$$\hat{y}_0 = 35 + 1.5(12) = 53$$

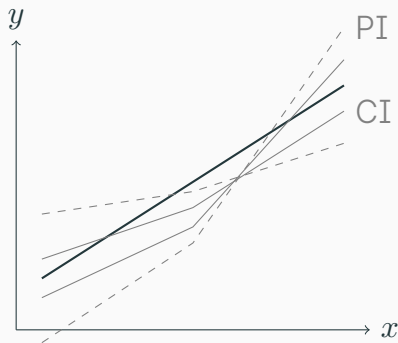
$$se(\text{mean}) = \sqrt{2.778 \left(\frac{1}{20} + \frac{4}{200} \right)} = 0.441$$

$$se(\text{pred}) = \sqrt{2.778 \left(1 + \frac{1}{20} + \frac{4}{200} \right)} = 1.724$$

- 95% CI for the mean response: $53 \pm 2.101(0.441) = (52.1, 53.9)$
- 95% PI for a new observation: $53 \pm 2.101(1.724) = (49.4, 56.6)$

CI Band vs. PI Band

- Both bands are narrowest at $x = \bar{x}$ and flare outward
- The PI band is always the wider of the two, everywhere
- Neither band should be trusted far outside the range of the observed x_i



Extrapolation Warning

- Both intervals are derived under the model fit *within* the observed range of x
- Predicting far outside that range assumes the linear relationship continues to hold — often unjustified

Question: in the advertising example (x ranging roughly 5–15), how much would you trust a prediction at $x_0 = 40$?

Summary

Putting It Together

- $\hat{\beta}_0, \hat{\beta}_1$ are normal; standardized by their standard errors they become t_{n-2}
- t -tests and CIs for β_0, β_1 follow the same template throughout
- ANOVA partitions $SST = SSR + SSE$; the F -test for $\beta_1 = 0$ is equivalent to the t -test in SLR
- $R^2 = SSR/SST = r^2$ measures explained variability, not model correctness
- Mean-response CIs are narrower than new-observation PIs — know which one you need

Next time: multiple linear regression

Questions?
